Rathod Adesh Siddhartha

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Profile

Master's student in Software Engineering at Blekinge Institute of Technology, building AI-agent systems whose output engineers can **trace, audit and review**. Built GateKeep, an agent harness that checks coding-agent work against **evidence** from real test runs and human approval gates, and CarbonLenz AI, which extracts evidence from messy documents, returns null rather than guess, and leaves decisions to explainable rules. Interested in applying **LLM agents and retrieval** to **root cause analysis** where uncertainty must stay visible. Available from **spring 2027**.

Education

M.Sc. in Software Engineering, Blekinge Institute of Technology, Karlskrona, Sweden Expected June 2027
B.Tech. in Computer Science and Engineering, JNTU Hyderabad, India Graduated 2025

Technical Skills

- **AI agents and LLMs:** agent orchestration (LangGraph, LangChain), MCP server design, LLM APIs (Gemini API), structured outputs, retrieval design (knowledge graph vs. RAG), prompt engineering
- **Traceability and human oversight:** audit trails, approval gates, reviewer UIs, rejection-with-feedback, log parsing, test-failure triage
- **Research and evaluation:** controlled-experiment design, hypothesis-test selection (t-test, Mann–Whitney U), effect sizes, threats to validity, research ethics
- **Languages and frameworks:** Python, JavaScript, C++, Node.js, FastAPI, React, PostgreSQL
- **Engineering:** software architecture, REST APIs, Git, Docker, CI/CD, pytest, scikit-learn

Projects

GateKeep: Test-Gated Agent Harness for AI-Generated Code GitHub

- Made agent-reported results verifiable, as measured by 100% of pass/fail outcomes coming from container exit codes across 68 passing tests, by parsing pytest logs from an air-gapped Docker sandbox and presenting them to human reviewers as evidence.
- Orchestrated a three-phase agent workflow (requirements, architecture, code) under human control, as measured by 20 governed MCP tools and 3 approval gates, by enforcing phase permissions in a server-side policy engine.
- Made every run traceable, as measured by prompts, diffs and test results chained in a SHA-256 audit ledger, by building a real-time review dashboard. Validated on pallets/itsdangerous. *Tech:* Python, FastAPI, LangGraph, MCP, PostgreSQL, Docker, React.

CI Test-Failure Triage: Log Parsing and Fault Classification In progress

- Attributed each CI failure to its likely cause (product, test, infrastructure, external library), as measured by precision, recall and macro-F1 on a hand-labelled set, by parsing raw build logs from public open-source projects into structured failure records and training a TF-IDF and logistic-regression baseline.
- *Tech:* Python, scikit-learn, regex log parsing, JSON.

Research

AI-Assisted Unit Test Generation in Safety-Critical Systems BTH, Mar 2026, team of 4

- Wrote a controlled-experiment proposal (research methodology course; not conducted) with 30–40 planned human participants: GitHub Copilot vs. manual CppUnit testing on a bug-seeded PX4 Autopilot C++ module.
- Defined 3 measurable outcomes (completion time, test error rate, seeded-bug detection) with pre-specified hypotheses, statistical tests and effect sizes; mapped threats to validity and research ethics.

Experience

Regulatory & Systems Analyst, CarbonLenz AI Karlskrona / Lund, Sweden | Jun–Aug 2026
Leapfrogs Program, Lund University Innovation

- Made supplier-evidence review explainable and traceable, as measured by a 0–100 risk score that flags the specific missing Annex VI criterion with a remediation step, by translating Regulation (EU) 2023/956 and Annex VI into 12 deterministic, testable verification rules.
- Cut review of a supplier certificate from an estimated 3–4 hours of manual compliance work to under 5 seconds, as measured on benchmark runs, by extracting 12 fields into schema-validated JSON with Gemini structured outputs and OCR, returning null instead of a guess when evidence is missing.
- Kept unverifiable AI output out of financial figures, as measured by every cost projection (2026–2034) being computed by a deterministic formula rather than the model, by separating LLM evidence extraction from a rules and cost engine.
- Kept exact legal values retrievable and auditable, as measured by each default intensity, phase-in factor (2.5% to 100%) and threshold resolving to one named knowledge node, by choosing a structured knowledge-graph retrieval design (Google OKF) over vector-similarity RAG. *Tech:* React, Node.js, Express, Gemini API, OCR.